Revisiting the Power of Prior: A Reproduction and Error Analysis of Training-Free Open-Vocabulary Segmentation with LLaVA

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Abstract

FSeg-LLaVA [Zhang et al., 2026] segments images without any segmentation training: a frozen LLaVA model decides which classes are present and provides a coarse spatial prior for each, and the Segment Anything Model (SAM) turns that prior into masks. We reimplement the method independently in Hugging Face Transformers, check it against the authors' released code, and analyse where it succeeds and fails. On all 1,449 PASCAL VOC 2012 validation images our implementation obtains 65.85 mIoU (95% CI [64.1, 67.3]); the released code, run in its own pinned environment, obtains 66.01, and the paper reports 68.0. The 2-point gap to the paper is therefore not specific to our port. Without retuning, the method transfers to LLaVA-1.6 Vicuna (65.02), LLaVA-1.6 Mistral (66.29) and COCO-Object (39.99 on 5,000 images), but not to Qwen2.5-VL (22.2 on a tuning split). Three diagnostics locate the remaining error. (i) *Recognition*: giving the pipeline the true class list adds +6.32 points on VOC and +10.42 on COCO. (ii) *Prior coverage*: when an object fills the frame, LLaVA's map covers only part of it; person is the weakest VOC class (19 IoU). (iii) *Depth*: middle layers give small but precise seeds that suit SAM, while deeper layers give better stand-alone maps but worse prompts. We also report eleven improvement attempts; none transferred from the tuning split to validation.

1 Introduction

Open-vocabulary semantic segmentation assigns every pixel to one of a set of class names chosen at test time. Most strong systems are trained on pixel-level labels or on large image–text corpora with dense objectives [Li et al., 2022, Ghiasi et al., 2022, Liang et al., 2023, Xu et al., 2022, Cha et al., 2023]. A training-free alternative asks whether the semantic and spatial knowledge already present in frozen models is enough. FSeg-LLaVA [Zhang et al., 2026] answers this with two frozen models: LLaVA [Liu et al., 2023, 2024a] supplies *what* is in the image and a coarse *where*, and SAM [Kirillov et al., 2023] supplies precise boundaries (Figure 1). The paper reports 68.0 mIoU on PASCAL VOC with no segmentation training.

A training-free pipeline has many coupled design choices (token positions, layers, thresholds, prompt construction), and a single final score cannot show which of them carry the result. We therefore reproduce the method and then take its error apart, guided by four questions:

Q1. Does the published result reproduce in an independent implementation, and how close is the authors' own released code?

Q2. Which design decisions matter, and where does the released code differ from the paper text?

Project report, Indian Institute of Technology Jodhpur, 2026.

Answers in brief

Q1 Yes, up to a shared 2-point gap: ours 65.85, released code 66.01 (paired Δ +0.16, CI [-0.25, +0.58]), paper 68.0 (Section 5.1).

Q2 SAM adds 47 points over LLaVA's maps. Reading the class token one position early costs 17 points; this detail is in neither the paper nor an obvious place in the code (Table 1 and Section 5.2).

Q3 Transfers to LLaVA-1.6 (Vicuna, Mistral) and to COCO-Object without retuning; fails on Qwen2.5-VL, whose hidden states are strongly anisotropic (Section 5.3).

Q4 The main loss is object pixels left as background (30.0%), not confused classes (2.9%). Recognition costs +6.32 points on VOC and +10.42 on COCO (Section 6).

Q3. Does the prior transfer to other VLMs (LLaVA-1.6, Qwen2.5-VL) and to COCO-Object?

Q4. Is the remaining error caused by recognition (which classes), by the prior (where), or by SAM (boundaries)?

Contributions. This is a reproduction and analysis study. We do not propose a new segmentation method, and our implementation does not beat the original.

- **A checked reproduction.** An independent Transformers implementation that agrees with the released code per image (Pearson $r = 0.97$), plus three implementation facts that are needed to reproduce the result but are not in the paper text (Table 1).
- **A decomposition of the error** into recognition, prior coverage and refinement, using oracle-recognition and maps-only controls on VOC and COCO, per class and per object size (Section 6).
- **Transfer results** for three further VLMs and a second dataset (Section 5.3), and a layer study that separates map quality from prompt quality (Section 5.4).
- **Negative results.** Eleven improvement attempts, chosen on a separate tuning split, none of which transferred to validation (Section 7).

2 Related work

Open-vocabulary segmentation with training. LSeg [Li et al., 2022], OpenSeg [Ghiasi et al., 2022] and OVSeg [Liang et al., 2023] align pixel or region features with text embeddings using segmentation or grounding supervision. GroupViT [Xu et al., 2022] and TCL [Cha et al., 2023] learn grouping from image–text pairs only. All of them train a dense head; FSeg-LLaVA trains nothing.

Training-free segmentation from CLIP. MaskCLIP [Zhou et al., 2022], SCLIP [Wang et al., 2024] and ProxyCLIP [Lan et al., 2024] modify the attention of a frozen CLIP [Radford et al., 2021] vision encoder to obtain dense patch–text similarities, relying on the contrastive embedding space. FSeg-LLaVA instead reads spatial evidence from the *language model* inside a multimodal LLM, and uses SAM for boundaries.

Multimodal LLMs for segmentation. LISA [Lai et al., 2024] and related models add a mask decoder to an MLLM and fine-tune it for segmentation. FSeg-LLaVA keeps LLaVA frozen and treats its hidden states as a prior, which also makes it a probe of how much spatial information an instruction-tuned VLM retains internally.

Reproducibility. Independent re-implementations often reveal undocumented choices that matter [Pineau et al., 2021]. We compare paper text, released code and our implementation under one protocol and one evaluator.

3 Background: the FSeg-LLaVA pipeline

Let I be an image and $\mathcal{C}$ a list of candidate class names (20 for VOC, 80 for COCO). The pipeline has four steps (Figure 1).

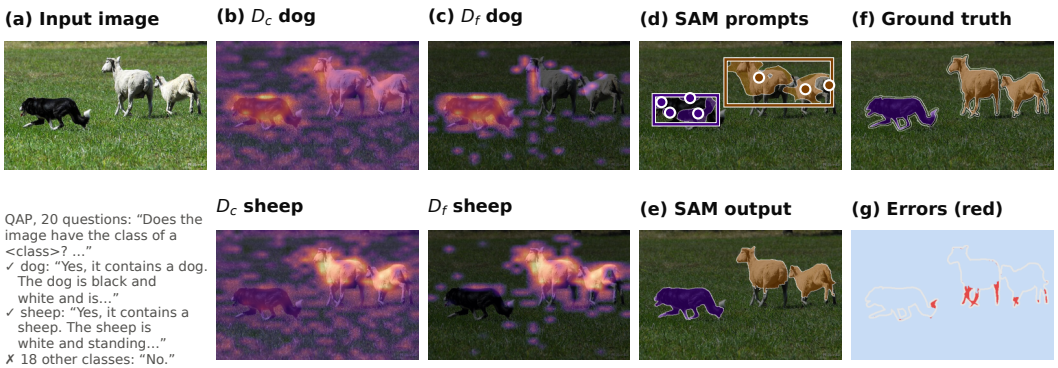

Figure 1: The FSeg-LLaVA pipeline on a real VOC validation image, with our implementation's intermediate outputs. (a) LLaVA is asked one yes/no question per class; classes answered "yes" are kept (verbatim answers shown). (b) The cosine similarity between the class-name token's hidden state and every image token (layers 7–13) gives a 24×24 response map D_c. (c) Thresholding and a union with description-to-image attention give the reliable map D_f. (d) Competition with a background level and CLIP prototypes gives a coarse mask, from which SAM receives positive points (dots), a box, and the other classes' points as negatives. (e) SAM's output after three refinement rounds. (f) Ground truth. (g) Pixel errors. The image was chosen by a fixed rule (two GT classes, both recognised, per-image mIoU in the 70–80th percentile).

Step 1: question–answer prompting (QAP). For each $c \in \mathcal{C}$, LLaVA receives the image and the query "*Does the image have the class of a c? Answer yes or no for this question. If no, just say 'No.' If yes, the answer has to be two sentences, the first sentence is 'yes, it contains a c.' the second sentence is to describe the c.*" A class is kept if the greedy answer contains "yes". A "yes" answer also produces the class-name token and a short description, which the next two steps use.

Step 2: text–visual response. Let h_c^l be the hidden state of the class-name token in the answer at language-model layer l, and h_i^l the hidden state of image token i (576 tokens, a 24×24 grid for LLaVA-1.5). The response map averages cosine similarities over a layer set L (hidden-state indices 7–13 in the released code):

$$D_c(i) = \frac{1}{|L|} \sum_{l \in L} \frac{\langle h_c^l, h_i^l \rangle}{\|h_c^l\| \, \|h_i^l\|}. \tag{1}$$

Step 3: reliable map and background. D_c is thresholded at $\tau_c = \min(\alpha \max_i D_c(i), \beta)$ with $\alpha = 0.75$ and $\beta = 0.19$. Separately, the attention from the description tokens to the image tokens is averaged over the 30 lowest-entropy (layer, head) pairs and binarised at its mean, giving M_A. The reliable map D_f keeps D_c where the threshold mask *or* M_A is on. Each pixel then goes to $\arg\max_c D_f$ if that value exceeds a global background level $\min(\alpha \max_{c,i} D_f, \beta)$, and to background otherwise.

Step 4: prototype refinement and SAM. CLIP patch features (layer -2) are averaged inside each class region to form prototypes, and pixels whose nearest prototype disagrees are removed. The refined region of each class yields positive points (centres of eroded components), a bounding box, and, as negative points, the positive points of the other classes. SAM ViT-H is prompted three times, each time also receiving its previous mask logits. Overlapping masks fall back to LLaVA's argmax map.

4 Experimental setup

Our implementation. We use Hugging Face Transformers checkpoints (`llava-hf/llava-1.5-7b-hf`, fp16) and the official SAM ViT-H weights. The code has three separate stages. *Trace* runs LLaVA once per image and caches the answers, $P(\text{yes})$, per-layer response maps, attention and CLIP features. *Segment* reruns only the SAM stage, so each ablation costs minutes instead of hours. *Evaluate* accumulates a confusion matrix. Every setting in this report is a configuration of the same code. Table 1 lists what we had to learn from the released code;

Table 1: Implementation facts needed to reproduce FSeg-LLaVA. The first two are where the released code differs from the paper text; the last is an indexing detail stated in neither. The last column is the change in mIoU when the paper-text (or wrong) version is used instead, measured with our implementation.

Component	Paper text	Released code (used here)	Δ mIoU
Combining D_c with M_A	product $D_c \cdot M_A$	union of threshold mask and M_A	−4.23 (val)
SAM refinement	single prompt	3 rounds, previous mask logits fed back	−3.01 (tuning)
Class-token hidden state	not specified	read *at* the class-name token, not one position earlier (a literal port of the `generate()` indexing)	−17.27 (val)

Table 2: Datasets and their role. Val sets are used only for final numbers; every choice was made on the tuning split.

Dataset	Split	Images	Classes	Role	Labels
PASCAL VOC 2012	`train`, seed-0 sample	200	20 + bg	tuning only	pixel masks
PASCAL VOC 2012	`val`	1,449	20 + bg	final results	pixel masks
COCO-Object	`val2017`	5,000	80 + bg	transfer	painted instance masks

the class-token position was the hardest to find (our first port read the state one position early and scored 48.6).

Datasets. Table 2 summarises the data. **PASCAL VOC 2012** [Everingham et al., 2010] has 20 object classes plus background, with pixel-level masks drawn by the VOC annotators; object boundaries are marked as a 5-pixel *void* band (label 255) that is ignored in scoring. We use the official segmentation `val` split (archive SHA-256 verified) only for final numbers. All design choices were made on a separate **tuning split** of 200 images drawn from VOC `train` with seed 0, which shares no images with val. **COCO-Object** is the standard training-free benchmark derived from COCO 2017 [Lin et al., 2014] `val`: the 80 "thing" categories plus background. We build it from the official instance annotations by painting every instance mask, crowd regions included, in annotation order; pixels with no instance are background. This follows GroupViT/TCL/SCLIP-style evaluations, but conversion scripts differ slightly across papers, so COCO numbers from different papers are not strictly comparable.

Models. LLaVA-1.5-7B [Liu et al., 2024a] (the paper's model), LLaVA-1.6 (LLaVA-NeXT [Liu et al., 2024b]) with Vicuna-7B [Chiang et al., 2023] and Mistral-7B [Jiang et al., 2023] language models, and Qwen2.5-VL-7B [Bai et al., 2025], all frozen. LLaVA-1.6 uses its own chat templates and only the base-resolution 24×24 image tokens for the maps. For Mistral, β was set to 0.16 on the tuning split (Appendix C). SAM is ViT-H [Kirillov et al., 2023] in all runs.

Protocol and metric. We report VOC21 mIoU (VOC) or 81-class mIoU (COCO). Intersections and unions are accumulated over the whole split, as in mmsegmentation, and void pixels are ignored. Confidence intervals are 95% percentile bootstrap intervals over images (1,000 resamples); paired differences resample the same images for both systems [Efron and Tibshirani, 1994]. Two *diagnostic controls* are never presented as results: *maps only* scores LLaVA's coarse label map without SAM, and *oracle recognition* replaces the QAP answers with the ground-truth class list and keeps everything else fixed. The released code was run in its pinned environment (PyTorch 2.1.2, Transformers 4.37, original `liuhaotian/llava-v1.5-7b` weights) on the same images and scored by our evaluator. All runs used one NVIDIA A30 (24 GB); compute is in Appendix E.

5 Results

5.1 Reproduction (Q1)

Our implementation obtains **65.85** mIoU on VOC val (CI [64.1, 67.3]) and the released code obtains **66.01** (CI [64.3, 67.5]; Table 3, plotted in Figure 10). The paired difference is +0.16 points (CI

Table 3: Main results (mIoU, 95% bootstrap CI in grey). "Maps only" and "Oracle recognition" are diagnostic controls, not methods. Δ is the paired difference to our LLaVA-1.5 reproduction on the same images; no LLaVA difference on VOC is significant. †With $\beta = 0.16$ (Appendix C).

System	VLM	*diagnostic* Maps only	Full pipeline	*diagnostic* Oracle rec.	Δ vs. ours
PASCAL VOC 2012 val (1,449 images)					
Paper [Zhang et al., 2026]	LLaVA-1.5-7B	–	68.0	–	–
Released code	LLaVA-1.5-7B	–	66.01 [64.3, 67.5]	–	+0.16 [-0.25, +0.58]
Ours	LLaVA-1.5-7B	18.72	**65.85** [64.1, 67.3]	72.16	–
Ours	LLaVA-1.6 Vicuna	16.45	65.02 [63.4, 66.8]	69.46	-0.82 [-2.32, +0.56]
Ours†	LLaVA-1.6 Mistral	19.90	66.29 [64.5, 67.9]	70.00	+0.44 [-1.19, +2.05]
COCO-Object val2017 (5,000 images)					
Ours	LLaVA-1.5-7B	18.40	**39.99** [38.8, 41.0]	50.41	–

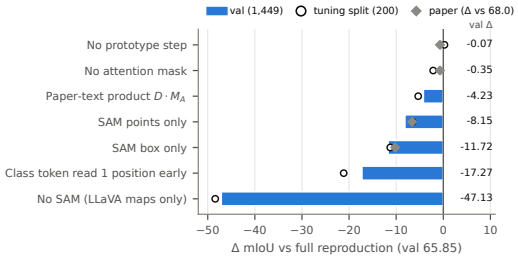

Figure 2: Ablations (Δ vs. the full reproduction). Bars: VOC val. Circles: tuning split. Diamonds: the paper's ablations relative to its 68.0. The trends match the paper (box + points > points > box; attention helps; prototypes matter little).

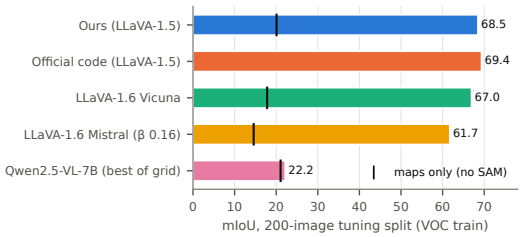

Figure 3: Five systems on the 200-image tuning split. Bars: full pipeline. Ticks: maps only. All LLaVA variants work with the paper's settings (Mistral with $\beta = 0.16$); Qwen2.5-VL does not, even after an 18-setting (α, β) grid and hidden-state centring.

[-0.25, +0.58]), so the two implementations are statistically indistinguishable. Per image they also agree closely: Pearson $r = 0.97$, and 89% of images are within 5 points (Figure 11, Appendix). Both are about 2 points below the paper's 68.0. Because the authors' own code in its pinned environment also lands at 66.0, the gap is not caused by our port, by the Hugging Face checkpoint conversion, or by newer library versions. One explanation we cannot test is the class-name file used for the paper, which is not public: on the tuning split, natural names ("potted plant" instead of "pottedplant") cost 1.4 points, so the vocabulary alone moves results by this order.

5.2 What matters inside the pipeline (Q2)

Figure 2 reproduces the direction of every ablation in the paper. SAM is the largest single component: the coarse LLaVA map alone scores 18.7, and SAM with points and a box adds 47 points. Points and box are complementary (points only -8.2, box only -11.7). The attention mask (-0.35 without it) and the prototype step ($+0.07$ without it) matter much less on val than in the paper (-0.7 each). Two choices from Table 1 matter far more than any listed in the paper: the union-versus-product combination (-4.2) and, above all, the class-token position (-17.3). The response map is sensitive to which token's state is read; the neighbouring token ("a") carries a generic, non-localised representation.

5.3 Transfer to other VLMs and to COCO (Q3)

LLaVA-1.6. Both LLaVA-1.6 models work without retuning (Table 3): Vicuna 65.02 (paired Δ vs. LLaVA-1.5 -0.82) and Mistral 66.29 (Δ +0.44); neither difference is significant. The *per-class* picture differs much more than the averages (Figure 4). Mistral more than doubles **person** IoU (40 vs. 19), the weakest class for LLaVA-1.5, and improves chair and sofa, but loses on boat, TV/monitor and aeroplane. Its recognition is more conservative (precision 0.952, recall 0.800, vs. 0.822/0.949 for LLaVA-1.5; Figure 6b): it removes false "yes" answers but misses more present classes.

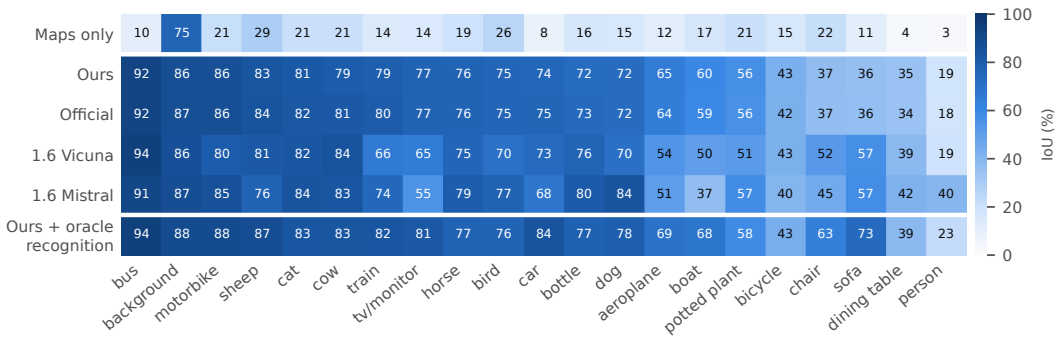

Figure 4: Per-class IoU on VOC val (classes sorted by our IoU). The two LLaVA-1.5 implementations are almost identical per class. LLaVA-1.6 Mistral gains on person, chair and sofa and loses on boat and TV/monitor. Person stays the weakest class for LLaVA-1.5 even with oracle recognition (23), so its problem is the prior, not recognition.

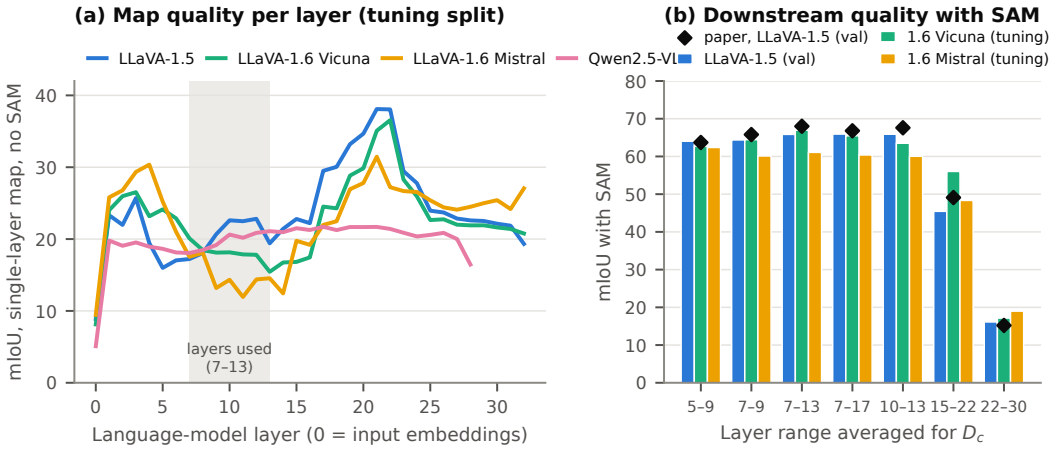

Figure 5: Layer study. (a) mIoU of the maps from a single layer, without SAM, for four VLMs (tuning split). Stand-alone map quality peaks at layers 20–22 for LLaVA-1.5 and LLaVA-1.6 Vicuna. (b) With SAM, the paper's layer ranges reproduce on val (diamonds: the paper's Table 3). Middle ranges are best, and the deep ranges that give the best maps collapse. Per-layer maps of one image are in Figure 13.

Qwen2.5-VL. With the same procedure, Qwen2.5-VL reaches only 22.2 on the tuning split, at the level of its maps alone (Figure 3). Its image-token hidden states are strongly anisotropic: the median cosine between class and image tokens is 0.47, against 0.09 for LLaVA-1.5, so every location looks similar to every class. Removing the mean image-token state did not help (best 17.0), and neither did an 18-setting (α, β) grid. The mechanism, as specified, depends on the LLaVA family's representation geometry; we do not claim it fails for every other architecture.

COCO-Object. On 5,000 COCO images with 80 classes, the pipeline reaches 39.99 (CI [38.8, 41.0]); SAM adds 21.6 points over the maps alone. Recognition is the larger limit here: the oracle class list adds +10.42 points (CI [+9.46, +11.19]), against +6.32 on VOC. Per class, the oracle gain tracks low recognition *precision* (Pearson $r = -0.62$; with recall $r = -0.21$; Figure 12), so on COCO the oracle mostly removes false "yes" answers. Toaster is the extreme case: LLaVA affirmed it 77 times for 8 true occurrences.

5.4 Where the spatial information sits: a layer study

The layer used to read D_c matters, and the best layer depends on what the map is used for (Figure 5). As stand-alone segmentations, single-layer maps improve sharply after layer 16 and peak at layers 21–22 (37.5 on val for LLaVA-1.5). As SAM prompts, the same deep ranges are the worst: 15–22 gives 45.5 and 22–30 gives 16.1, against 65.9 for 7–13. The reason is seed precision. Middle-layer maps mark only 5% of pixels, and 79% of those are correct, so SAM receives reliable points inside

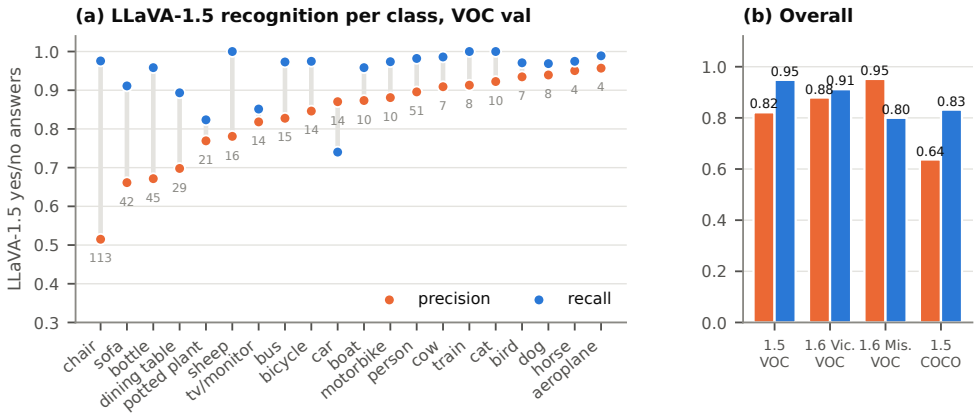

Figure 6: Recognition (QAP) errors. (a) Per-class precision and recall of LLaVA-1.5's yes/no answers on VOC val; grey numbers count false "yes" answers. Precision is low for chair, dining table, sofa and bottle, which co-occur with other furniture and indoor scenes. (b) Overall precision and recall per model and dataset.

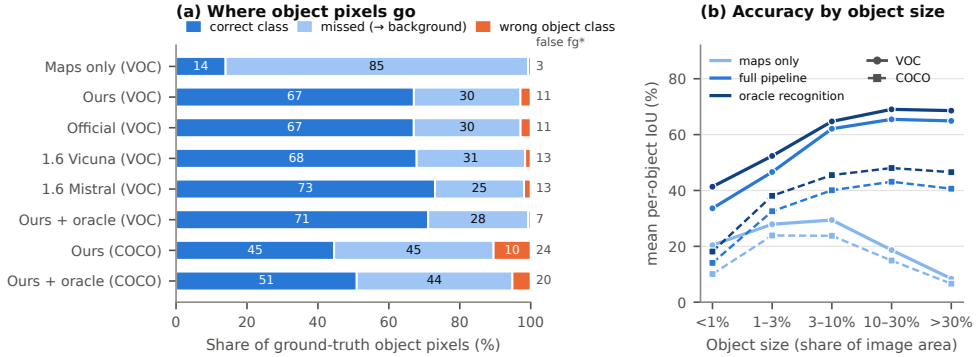

Figure 7: Where the error comes from. (a) Fate of every ground-truth object pixel: correctly labelled, left as background, or given another object class. The number to the right is the count of background pixels labelled as an object, per 100 object pixels. (b) Mean per-object IoU by object size (share of image area) for maps only, the full pipeline and oracle recognition, on VOC (solid) and COCO (dashed).

the object. Layers 15–22 mark larger, less precise regions (precision 0.50), and the resulting SAM masks cover 43% of pixels against 27% true foreground. Both LLaVA-1.6 models follow the same pattern; Qwen2.5-VL's curve is flat at about 20 for every layer, matching its failure above. **For training-free pipelines, layers should be chosen by the quality of the downstream prompts, not by how good the maps look.**

6 Error analysis (Q4)

Recognition. Over $1{,}449 \times 20$ VOC questions, LLaVA-1.5 answers "yes" wrongly 442 times and "no" wrongly 110 times (precision 0.822, recall 0.949; Figure 6). False "yes" answers concentrate on chair, dining table, sofa and bottle: in an indoor scene, LLaVA tends to affirm furniture that is plausible but absent. A false "yes" is costly, because the class then competes for pixels and is passed to SAM. On COCO, with 80 questions per image, false "yes" answers dominate: precision drops to 0.637 (6,740 false "yes" against 2,387 misses; recall 0.832).

Missed pixels, not confused classes. Figure 7a follows every ground-truth object pixel. With LLaVA-1.5 on VOC, 67.1% are labelled correctly, 30.0% are left as background, and only 2.9% receive another object class. Oracle recognition removes most of the wrong-class share (to 0.7%) but leaves the background share almost unchanged (28.2%). The dominant failure is therefore an *incomplete prior*: the object is recognised, but part of it is never prompted. On COCO the background share is larger (45.0%).

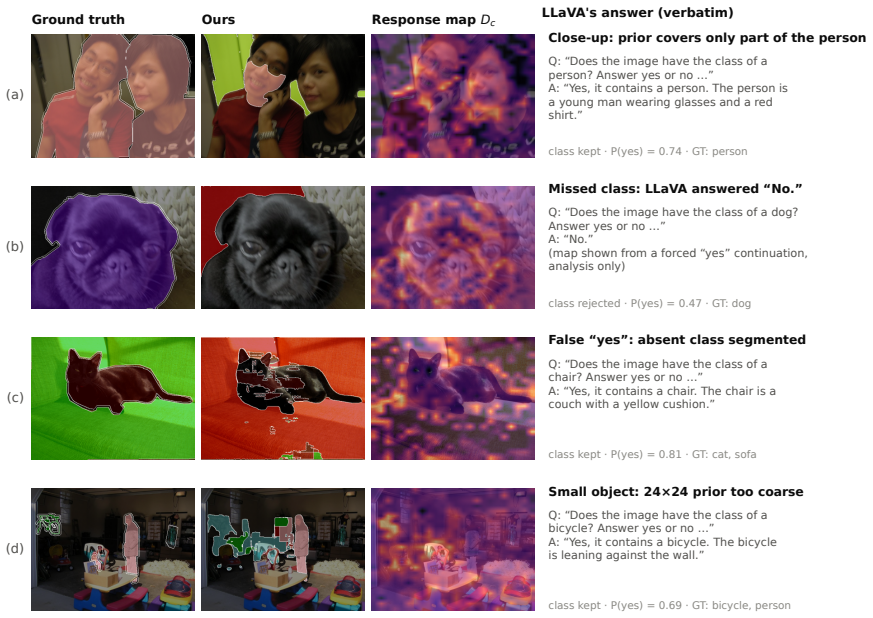

Figure 8: Failure types with LLaVA-1.5's verbatim answers, chosen by fixed rules (Appendix D). (a) Close-up: the prior and SAM cover one face. (b) Missed class: LLaVA answers "No." to a dog filling the frame ($P(\text{yes}) = 0.47$). (c) False "yes": LLaVA calls the sofa "a chair ... a couch with a yellow cushion", and the pipeline labels it chair. (d) A small bicycle that is recognised but falls between grid cells.

Object size. Accuracy rises with object size up to objects of 10–30% of the image and then flattens (Figure 7b). Small objects (<1% of the image; 32% of COCO objects) reach only 14.1 IoU on COCO (33.6 on VOC): on a 24×24 grid each token covers about 0.17% of the image, so a small object occupies one or two tokens and the prior often misses it. For large objects the maps alone collapse (8.4 IoU on VOC for objects above 30% of the image, against 64.9 with SAM): the response concentrates on the most distinctive part, such as a face or a wheel. SAM's box usually recovers the full extent; the close-ups where it does not are the person failures below.

Qualitative evidence. Figure 19 (Appendix) compares all systems on six images at fixed quantiles, and Figure 8 shows one example of each failure type with LLaVA's verbatim answer. (a) In close-ups, the person response covers the face or skin. Over all of VOC val, 78% of person pixels end up as background, and 84% in the 49 images where a person covers more than 45% of the frame. (b) A missed class has no map at all, so nothing can recover it. (c) A false "yes" produces a confident description of an object that is not there, and SAM then segments whatever region the map points to. (d) Small objects fall between grid cells. These four types correspond to the three quantitative findings above: recognition, prior coverage and resolution.

7 What did not work

We tried to fix each error source found above, choosing on the tuning split and keeping val untouched (Figure 9). For *recognition*, we tested an evidence filter that drops unsure "yes" classes with tiny maps, calibrated thresholds on $P(\text{yes})$, and a query alias ("person, people"). For the *incomplete prior*, we tested per-class background levels, relative backgrounds, a SAM box extended to the deep-layer region, and one SAM prompt per connected region. For *representation*, we tested map normalisation and hidden-state centring. Per-class back-

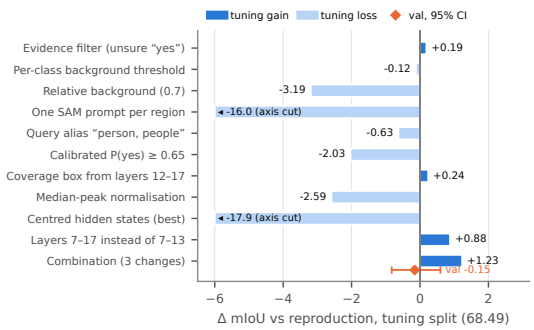

Figure 9: Improvement attempts, each aimed at a measured error source, on the tuning split. The only gain above noise (the combination, +1.23) was frozen and run once on val, where it scored -0.15 (CI [-0.83, +0.60]).

grounds raised person IoU by 13.8 on the tuning split but cost other classes as much. One prompt per region was much worse, because the LLaVA region is fragmented *within* a single object. The only candidate above noise was a combination of three small changes (+1.23 on tuning, CI $[-0.5, +3.5]$). We froze it and ran it once on val, where it scored -0.15 (CI [-0.83, +0.60]). The tuning gain was sampling noise from 200 images, so our final number is the plain reproduction.

8 Discussion and limitations

What the results support. (1) The method reproduces: two independent implementations agree per image, and the remaining gap to the paper is shared by the authors' code. (2) It transfers within the LLaVA family and to COCO, but not to Qwen2.5-VL as specified. (3) SAM does most of the boundary work, so the useful output of LLaVA is a *precise seed* rather than a complete map.

Limitations. We use one SAM variant and 7B-scale VLMs only. The Qwen2.5-VL result comes from the 200-image tuning split and one family of fixes, so it shows that the method does not transfer *as is*, not that it cannot be adapted. We did not evaluate PASCAL Context, ADE20K or COCO-Stuff, so we cannot speak to stuff classes or vocabularies of 150+ classes. The oracle numbers are diagnostic upper bounds, not methods. The layer-study and LLaVA-1.6 tuning numbers use 200 images, and differences below about 1.5 points there are within noise (Section 7).

Broader impact. Training-free segmentation makes masks for new classes cheap, but it inherits the VLM's recognition errors and biases. The confident false descriptions in Figure 8c show that an MLLM's "yes" is not evidence on its own; downstream users should not treat its masks as verified detections. We use public benchmarks only.

9 Conclusion

Our independent reproduction of FSeg-LLaVA (65.85 VOC mIoU) matches the released code (66.01), and both sit about 2 points below the published 68.0. The method transfers to LLaVA-1.6 and COCO-Object without retuning but not to Qwen2.5-VL. The remaining error comes from LLaVA's yes/no recognition (+6.32 points on VOC, +10.42 on COCO) and from priors that are incomplete on large objects and too coarse for small ones, while SAM carries the boundaries. For future training-free pipelines, our results suggest two concrete targets: better-calibrated recognition and layer choices made for prompt precision rather than map quality.

Acknowledgments and Disclosure of Funding

We thank the authors of FSeg-LLaVA for releasing their code, which made the side-by-side comparison possible, and IIT Jodhpur for access to its GPU cluster. No external funding supported this work; the authors declare no competing interests.

References

S. Bai, K. Chen, X. Liu, et al. Qwen2.5-VL technical report. *arXiv:2502.13923*, 2025.

J. Cha, J. Mun, and B. Roh. Learning to generate text-grounded mask for open-world semantic segmentation from only image-text pairs. In *CVPR*, 2023.

W.-L. Chiang, Z. Li, Z. Lin, et al. Vicuna: An open-source chatbot impressing GPT-4 with 90%* ChatGPT quality. LMSYS blog, 2023.

B. Efron and R. J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall/CRC, 1994.

M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman. The PASCAL visual object classes (VOC) challenge. *IJCV*, 88(2):303–338, 2010.

G. Ghiasi, X. Gu, Y. Cui, and T.-Y. Lin. Scaling open-vocabulary image segmentation with image-level labels. In *ECCV*, 2022.

A. Q. Jiang, A. Sablayrolles, A. Mensch, et al. Mistral 7B. *arXiv:2310.06825*, 2023.

A. Kirillov, E. Mintun, N. Ravi, et al. Segment anything. In *ICCV*, 2023.

X. Lai, Z. Tian, Y. Chen, Y. Li, Y. Yuan, S. Liu, and J. Jia. LISA: Reasoning segmentation via large language model. In *CVPR*, 2024.

M. Lan, C. Chen, Y. Ke, X. Wang, L. Feng, and W. Zhang. ProxyCLIP: Proxy attention improves CLIP for open-vocabulary segmentation. In *ECCV*, 2024.

B. Li, K. Q. Weinberger, S. Belongie, V. Koltun, and R. Ranftl. Language-driven semantic segmentation. In *ICLR*, 2022.

F. Liang, B. Wu, X. Dai, et al. Open-vocabulary semantic segmentation with mask-adapted CLIP. In *CVPR*, 2023.

T.-Y. Lin, M. Maire, S. Belongie, et al. Microsoft COCO: Common objects in context. In *ECCV*, 2014.

H. Liu, C. Li, Q. Wu, and Y. J. Lee. Visual instruction tuning. In *NeurIPS*, 2023.

H. Liu, C. Li, Y. Li, and Y. J. Lee. Improved baselines with visual instruction tuning. In *CVPR*, 2024.

H. Liu, C. Li, Y. Li, B. Li, Y. Zhang, S. Shen, and Y. J. Lee. LLaVA-NeXT: Improved reasoning, OCR, and world knowledge. Blog post, 2024.

J. Pineau, P. Vincent-Lamarre, K. Sinha, et al. Improving reproducibility in machine learning research. *JMLR*, 22(164):1–20, 2021.

A. Radford, J. W. Kim, C. Hallacy, et al. Learning transferable visual models from natural language supervision. In *ICML*, 2021.

F. Wang, J. Mei, and A. Yuille. SCLIP: Rethinking self-attention for dense vision-language inference. In *ECCV*, 2024.

J. Xu, S. De Mello, S. Liu, et al. GroupViT: Semantic segmentation emerges from text supervision. In *CVPR*, 2022.

Zhang et al. The power of prior: Training-free open-vocabulary semantic segmentation with LLaVA. In *CVPR*, 2026. Code: https://github.com/zbf1991/FSeg-LLaVA.

C. Zhou, C. C. Loy, and B. Dai. Extract free dense labels from CLIP. In *ECCV*, 2022.

Technical Appendices

A Additional results

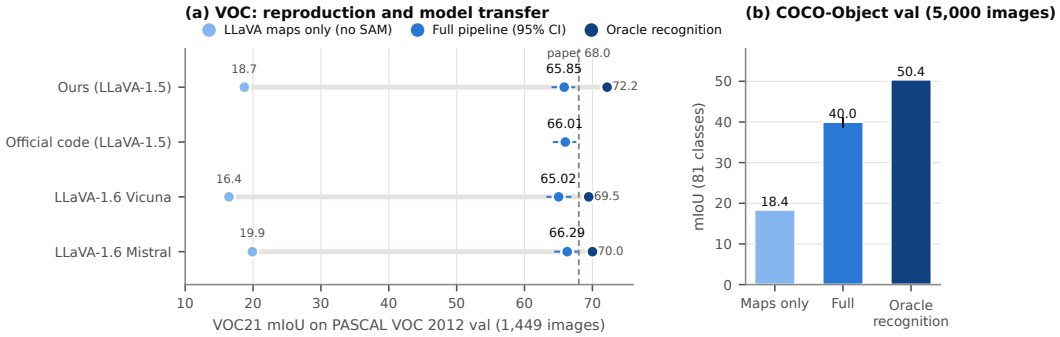

Figure 10: Main results with uncertainty. Each row shows LLaVA's maps without SAM (light), the full pipeline with its 95% bootstrap CI (mid), and the pipeline with the true class list (dark; diagnostic only). (a) VOC val; dashed line: the paper's 68.0. (b) COCO-Object val with LLaVA-1.5.

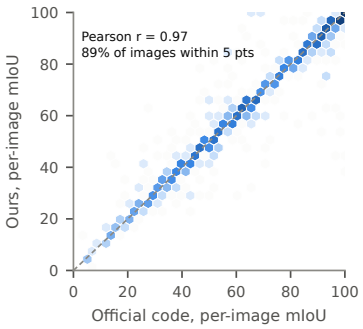

Figure 11: Per-image agreement between our implementation and the released code on the 1,449 VOC val images (hexbin counts; dashed line = equality). Pearson $r = 0.97$; 89% of images are within 5 points. Most remaining disagreement comes from images where one implementation keeps a borderline class and the other does not; different LLaVA weights (original vs. HF-converted) and fp16 kernels can flip such greedy answers.

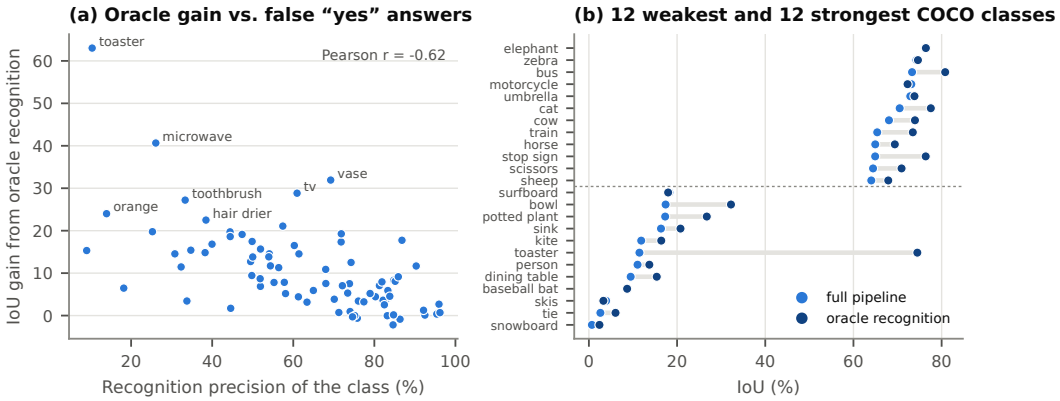

Figure 12: COCO-Object per class. (a) IoU gain from oracle recognition against the class's recognition precision: classes with many false "yes" answers gain most. (b) The 12 weakest and 12 strongest classes (full pipeline vs. oracle recognition).

B Additional qualitative results

This appendix collects further examples from the saved outputs of the runs in Table 3. As everywhere in the report, each image is chosen by a fixed rule stated in its caption (and logged in `data/selection.json`), not by hand. White numbers are per-image mIoU unless stated otherwise.

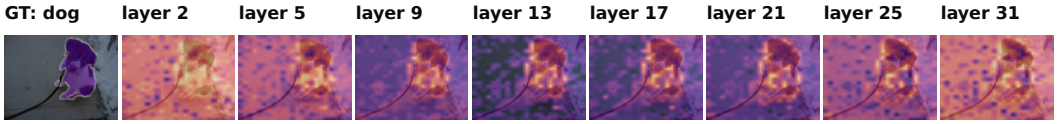

Figure 13: Response map D_c of a single layer, from early to late layers, for one VOC val image (selection rule in Appendix D). Each map is min–max scaled.

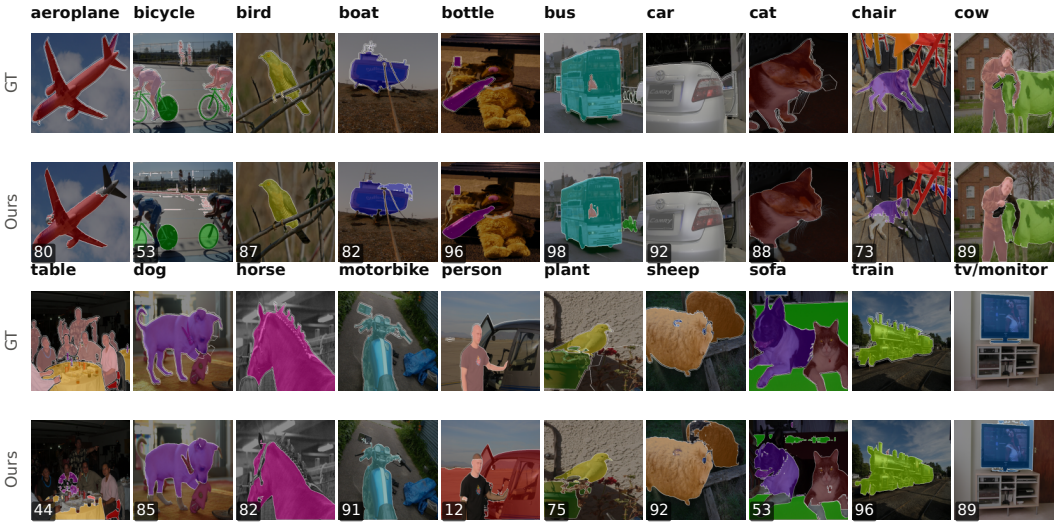

Figure 14: One example per VOC class (top: ground truth, bottom: our full pipeline). For each class we show the object with the *median* IoU among objects of that class covering 5–60% of the image, so each pair is a typical case for its class; the number is that object's IoU. Person (12) and table (44) are typical failures: the prior covers only the face or part of the table, which matches Figure 4.

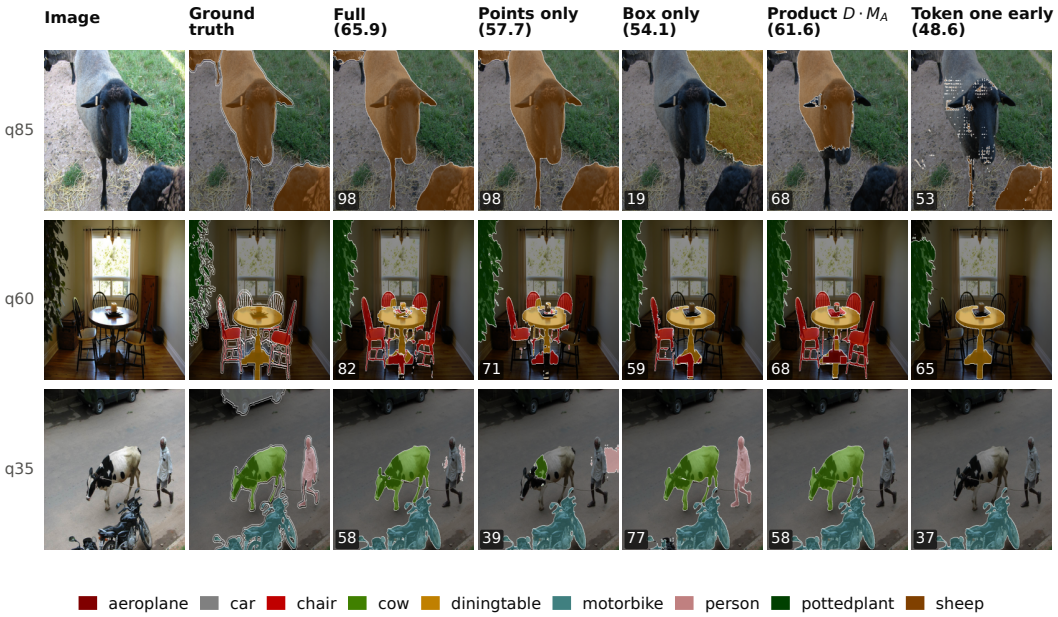

Figure 15: Ablations on the same images (rows at the 85th, 60th and 35th percentile of our per-image mIoU; column headers give each setting's mIoU on all of VOC val). Box-only prompts can grab the wrong extent (sheep, 19), and reading the class token one position early (Table 1) gives scattered, unlocalised seeds.

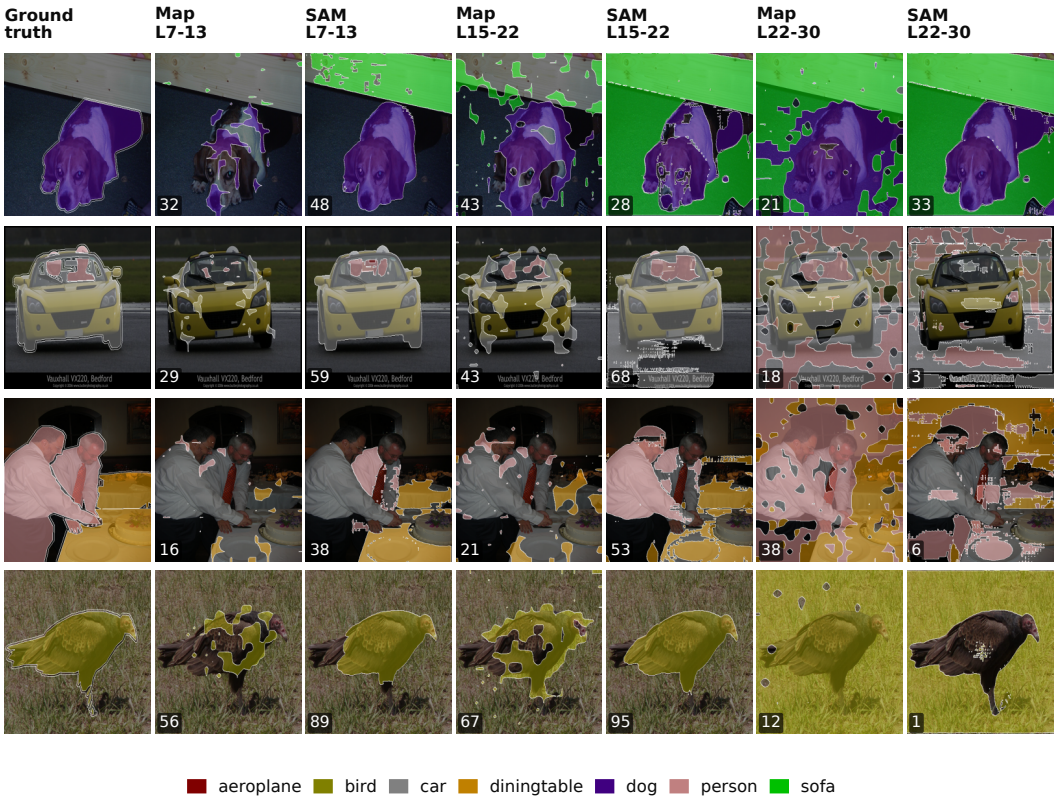

Figure 16: Layer ranges: LLaVA map vs. SAM output for dog, car, person and bird (per class, the median-IoU object covering 10–50% of the image). Deep-layer maps (L22–30) cover large, imprecise regions, so SAM receives wrong prompts and can collapse (car 3, bird 1); middle layers give smaller but cleaner seeds. Single images vary (here L15–22 wins on car and bird), which is why Figure 5 reports the aggregate over all of val.

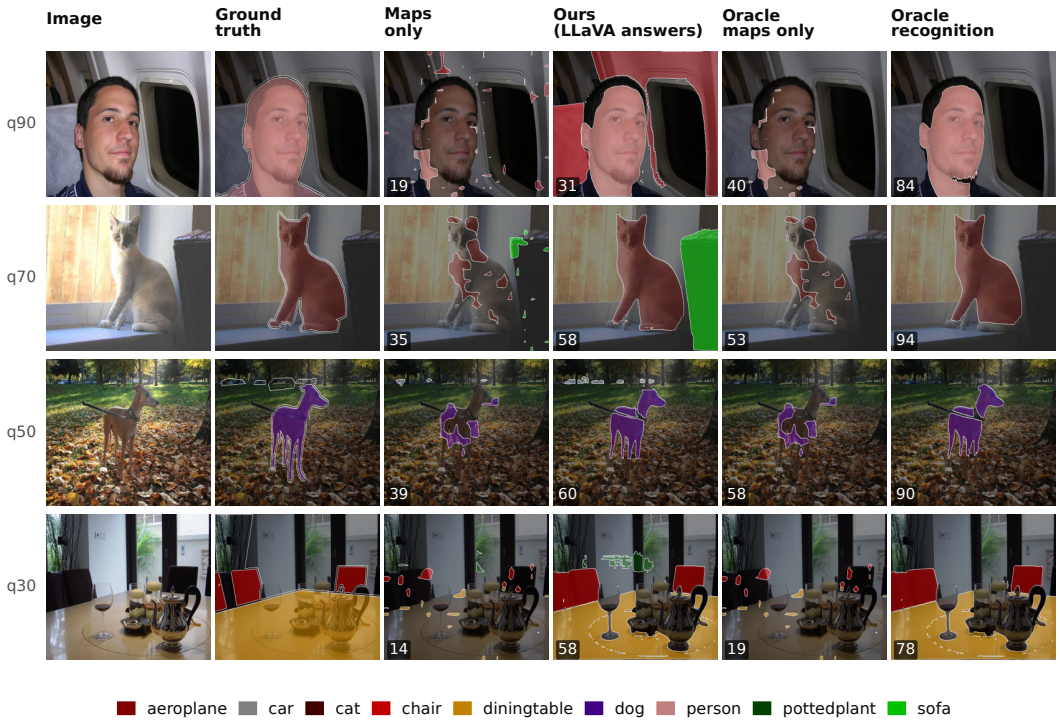

Figure 17: What oracle recognition fixes (images where it gains >5 points, at the 90th, 70th, 50th and 30th percentile of the gain). It removes falsely affirmed classes (the chair next to the face, the green region beside the cat) and restores rejected ones; it does not fix incomplete extent.

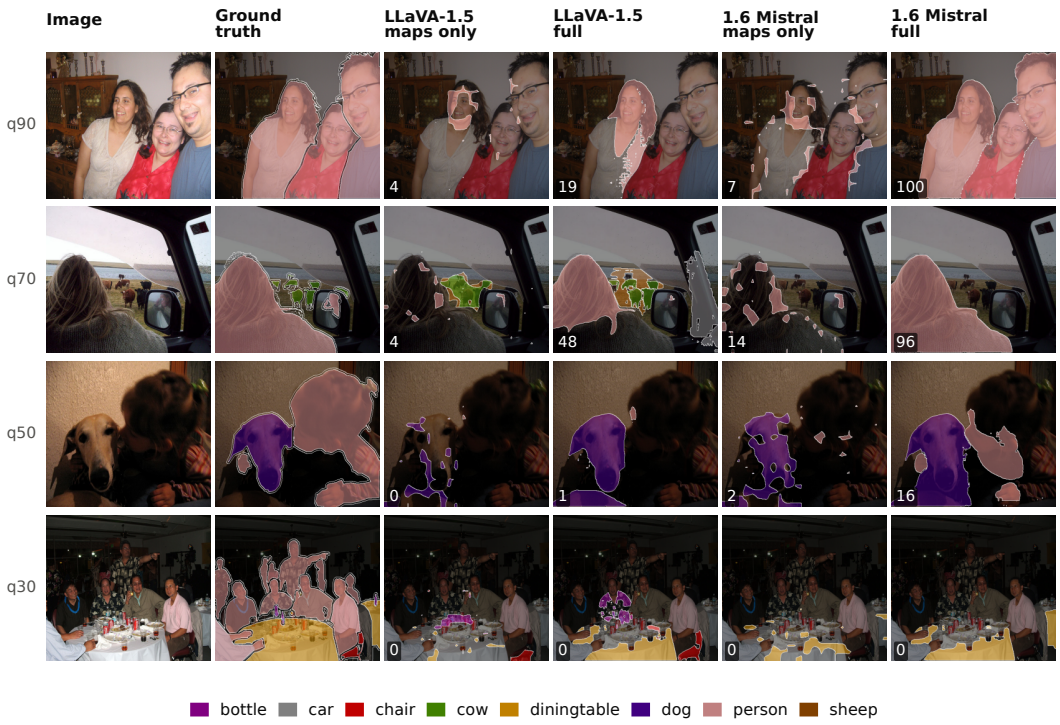

Figure 18: Large persons: LLaVA-1.5 vs. LLaVA-1.6 Mistral (person objects covering >30% of the frame, at the 90th, 70th, 50th and 30th percentile of Mistral's gain in person IoU). Numbers are *person* IoU. Mistral's map spreads over the body more often, which explains its higher person IoU (Figure 4); crowded scenes (bottom row) still fail for both.

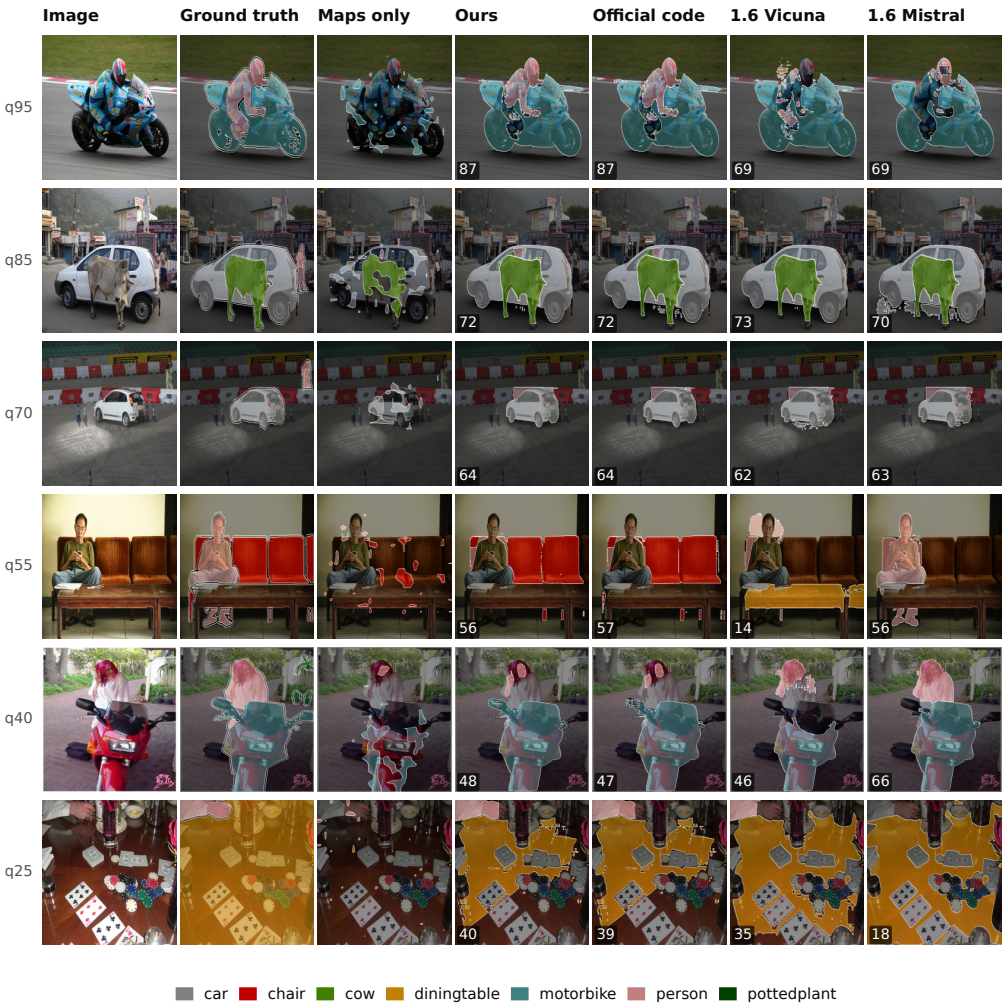

Figure 19: Qualitative comparison on VOC val. Rows are images at fixed quantiles (95th to 25th) of our per-image mIoU, among images with at least two classes, so the rows span good to below-median cases rather than hand-picked successes. Numbers: per-image mIoU. "LLaVA maps only" is the coarse prior before SAM.

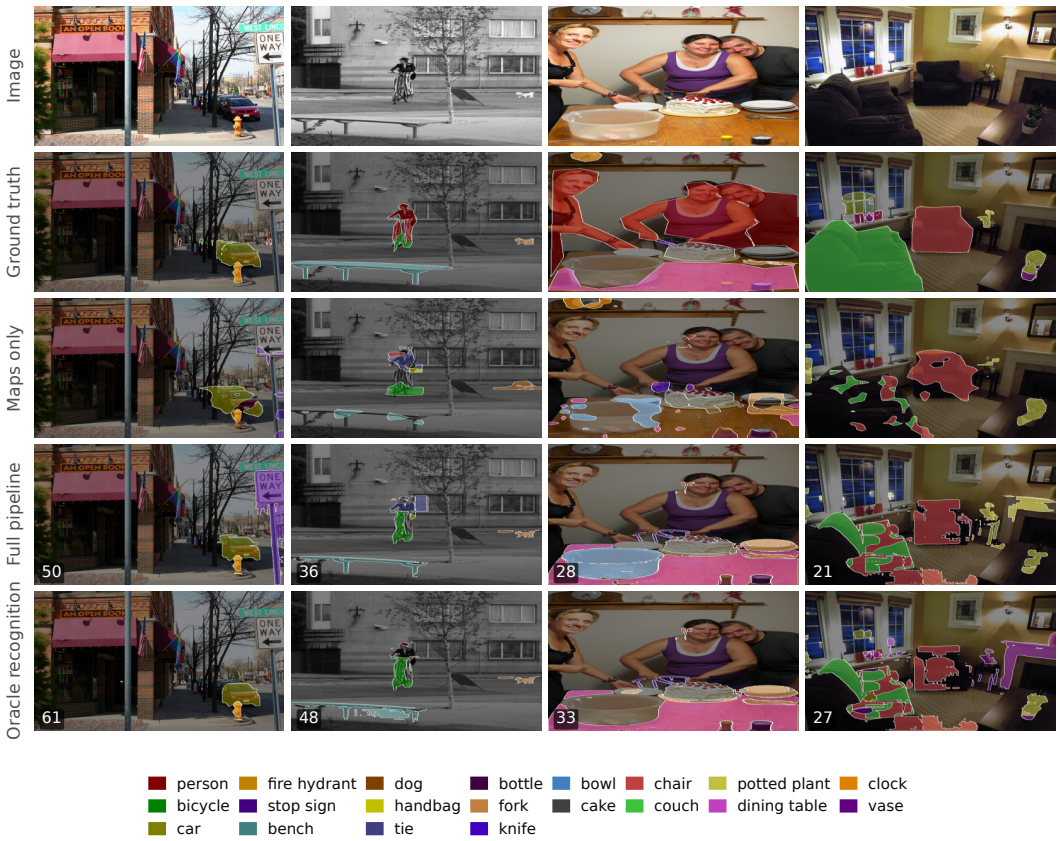

Figure 20: COCO-Object examples at the 90th, 70th, 50th and 30th percentile of per-image mIoU (images with $\geq$3 classes). Numbers: per-image mIoU. Small objects and crowded scenes lose most; oracle recognition restores missed classes but not missing extent.

C Tuning decisions

All decisions used the 200-image tuning split (Table 4). LLaVA-1.5 keeps the released defaults, which were also the best values in a local search. LLaVA-1.6 Vicuna uses the same values. LLaVA-1.6 Mistral produces lower cosine levels, so it uses $\beta = 0.16$ on val. Qwen2.5-VL was given a 6×3 (β, α) grid (best 22.2), all eight layer ranges (14.2–15.8 with SAM), three normalisations, and centred hidden states (best 17.0); its best setting is reported in Figure 3.

Table 4: Hyperparameter search on the tuning split (mIoU). Chosen values in bold.

Model	Parameter	Values	mIoU per value			
LLaVA-1.5	α	0.65 / **0.75** / 0.85	67.5	**68.5**	67.3	
LLaVA-1.5	β	0.16 / **0.19** / 0.22	65.0	**68.5**	64.3	
LLaVA-1.6 Mistral	β	0.13 / **0.16** / 0.19 / 0.22	53.0	**61.7**	61.1	61.3

D Example selection

To avoid cherry-picking, every image shown was chosen by a fixed rule, written in the figure script (`reports/neurips/make_figures.py`) and logged with the chosen IDs in `data/selection.json`.

- Figure 1: the best-ranked image with two GT classes, both recognised, and per-image mIoU in the 70–80th percentile.
- Figure 13: the 4th-best single-object animal image with 15–35% area and IoU ≥ 0.85.
- Figures 14 to 18: rules in their captions (Appendix B).
- Figures 19 and 20: fixed quantiles of per-image mIoU.
- Figure 8: (a) the median-IoU case among person-only images where the person covers >45% of the frame, is recognised, and has IoU <0.3; (b) the largest object whose class LLaVA rejected, with IoU <0.05, that oracle recognition recovers to ≥0.7; (c) the largest predicted area of an absent class (every third val image scanned); (d) the median case among recognised non-person objects below 2% of the image with IoU <0.2. For a rejected class the map is taken from a forced "yes" continuation that the pipeline itself never uses.

E Compute

Table 5: Wall-clock time of the main jobs on one NVIDIA A30 (24 GB). LLaVA tracing is the dominant cost; every SAM-only ablation then reuses the cache.

Job	Images	Time (h:mm)
LLaVA-1.5 trace, VOC val	1,449	2:49
SAM + scoring, 8 VOC val configurations	1,449	1:14
Layer study, VOC val (per-layer traces + 16 configurations)	1,449	6:44
Released code, VOC val (pinned environment)	1,449	3:06
LLaVA-1.6 Vicuna, trace + SAM, VOC val	1,449	4:19
LLaVA-1.6 Mistral, trace + SAM, VOC val	1,449	2:24
LLaVA-1.5, trace + SAM, COCO-Object val	5,000	13:46
All GPU jobs of the project (40, incl. cancelled reruns)		≈ 48:00